

Marcus Morales

9/30/26

1. My Understanding of the challenge

The goal of the challenge is to design, build, test, demonstrate, and document a device controlled by a digital system such as an Arduino or Raspberry Pi. The device needs to include functioning elements from all five required categories and have those parts work together as one complete system, NOT as a separate project. - A strong project would be something useful, reliable, creative, safe and well built as one piece. It needs to show research effort while following all material and cost limits.

2. Requirements and Constraints

- Control system: Use a Arduino or Raspberry Pi (which is reliable.)
Vex VS Brains are NOT Allowed.
- Display: Include LED's, a display, or other useful output.
- Manual User Input: Include something such as buttons, switches, joystick, Key pad, or touchscreen.
- Automatic sensor: Include a sensor that allows the device to respond automatically without direct user input.
- Materials: Up to 250g of PLA or PETG filament can be used.
- Up to \$20 in pre-approved student purchased parts per project
- Project needs an engineering notebook, poster, Github report / depositories, CAD files, wiring diagrams, commented code, and demonstration video.
- Sources: Any outside code or circuits must be understood, modified or integrated meaningfully, and properly cited.

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3. Personal Readiness

Skills and resources I already have.

- Experience with Vex V5 robotics
- Experience with electronics, circuits, sensors, and programming.
- Experience with 3D modeling and 3D printing
- Access to school lab equipment and materials.

Skills I may need to learn or improve

- Improve my ability to program better, and make multiple components communicate.
- Learn more about integrating hardware, software, and electronics into one complete system.
- Improve mechanical design so mechanisms are reliable and repeatable

4. Three Possible Directions

Idea and intended user / problem	output or display	Manual user input	Automatic sensor	Actuator, mechanism, or hardware	logic processing and control	Biggest risk or unknown
Idea 1: Smart Package locker helps protect packages by automatically detecting when a package is placed inside.	LCD showing Locker status	Keypad/button to enter a code.	Distance sensor detect a package.	Servo controlled locking mechanism.	Controller checks the code and sensor, then controls the lock.	Making the lock mechanism strong and reliable.
Idea 2: Automatic Basket ball hoop - Detects a basket ball and keeps track of successful shots.	LED scoreboard / display showing score.	Button to start / reset the game.	Distance sensor detects the ball.	Servo controlled ball return mechanism.	Controller detects shots, counts points, and activates the ball return.	Getting the ball detection and return mechanism to work consistently.
Idea 3: Smart Recycling Sorter helps separate recyclable waste into different containers.	LED / display shows the detected category.	Button to start the sorting process.	Distance / color sensor detects the object.	Servo-controlled sorting gate.	Controller processes the sensor reading and moves the object into the correct section.	Getting the sensor to distinguish objects consistently.

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5. Current Front-Runner

My current front runner would be the Smart Package Locker because it has a useful purpose and can combine all five functional categories into one system. It could use a keypad, LCD, Distance sensor, servo lock, and programmed controller. It would also require mechanical Design and testing to make the locking system reliable.

6. Anticipated Challenges

1. Challenge: Designing the locking mechanism
Why it matters: The ~~aperture~~ lock needs to securely close and reliably open when the correct code is entered.
Possible preparation or test: Build and test a small locking mechanism before making the final version.
2. Challenge: Detecting packages correctly
Why it matters: The system needs to know when a package has been placed inside the locker.
Possible preparation or test: Test different sensor positions and distances to find the most reliable setup.
3. Challenge: Integrating the electronics and programming
Why it matters: The keypad, sensor, display, servo, and controller all need to communicate and work together.
Possible preparation or test: Test each component individually before combining them into the complete system.

7. Individual or Partner Project

At this point in my idea, I lean towards working with a partner, as long as the partner is reliable and willing to communicate and contribute consistently. Having complementary skills could make it easier to work on mechanical construction, electronics, programming, and documentation.

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8. Questions and Clarifications

1. ~~What~~ How much of the final devices need to be completed by each major checkpoint?
2. Are there any restrictions on using features as Wi-Fi or Bluetooth?
3. How many partners per project or is it just a pair (2)?

9. Preparation Plan

- Research possible sensors and controllers that could be used for my project.
- Sketch several possible device designs and mechanisms using design matrices.
- Create small prototypes to test.
- Plan how each of the five functional categories will be integrated into the final device.
- Keep track of components and their costs and limit of 250g effort.

10. Final Takeaway

The most important principle is that I need to choose a project that can successfully integrate all five functional categories into one reliable system. A project should not be chosen just because it sounds interesting. It needs to be useful, reliable, effortable, scorable, and capable of demonstrating meaningful research and effort.

Assistance Acknowledgment

I partially used AI to help me plan and brainstorm possible ideas for this project. I also used it to help organize my thoughts and understand the project requirements, but the final ideas and decisions are on my own.

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